

CHI CHANG

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PROFESSIONAL EXPERIENCE

2024 – present	Associate Professor – Tenured <i>Office of Medical Education Research and Development</i> College of Human Medicine, Michigan State University, East Lansing, MI <i>Department of Epidemiology and Biostatistics</i> College of Human Medicine, Michigan State University, East Lansing, MI
2025 Fall	Visiting Scholar Stanford Prevention Research Center Department of Medicine College of Medicine, Stanford Medicine, Palo Alto, CA
2025 Fall	Visiting Scholar Department of Surgery College of Medicine, University of California, San Francisco, Mission Bay, CA
2018 – 2024	Assistant Professor – Tenure Track <i>Office of Medical Education Research and Development</i> College of Human Medicine, Michigan State University, East Lansing, MI <i>Department of Epidemiology and Biostatistics</i> College of Human Medicine, Michigan State University, East Lansing, MI
2022 – 2024	Research Scientist <i>Department of Veterans Affairs</i> Battle Creek VA Medical Center, Battle Creek, MI
2016 – 2017	Research Assistant Professor <i>Methodology Group</i> College of Education, The University of New Mexico, Albuquerque, NM <i>Department of Individual, Family, and Community Education</i> College of Education, The University of New Mexico, Albuquerque, NM

EDUCATION

2016	Michigan State University , East Lansing, MI Ph.D., Measurement & Quantitative Methods Major: Measurement; Minor: Statistics Department of Counseling, Educational Psychology, and Special Education Advisor: Dr. Kimberly S. Kelly
2016	Michigan State University , East Lansing, MI M.S., Biostatistics Department of Epidemiology and Biostatistics

2008	Advisor: Dr. Joseph Gardiner National Chengchi University, Taipei, Taiwan M.A., Educational Administration and Policy Department of Education Advisor: Dr. Hsen-Hsing Ma
2004	National Kaohsiung Normal University (NKNU), Kaohsiung, Taiwan B.A., Educational Psychology and Counseling Department of Education Advisor: Dr. Tsuey-Shing Fu

ACCREDITATIONS

2009	Ministry of Education, Taiwan Educational Administration Track, Senior Examination for the Civil Service Diplomat Certificate
2004	National Kaohsiung Normal University, Kaohsiung, Taiwan Student Teacher Elementary School Teaching Certificate
2004	National Kaohsiung Normal University, Kaohsiung, Taiwan Student Teacher Secondary School Teaching Certificate

CERTIFICATES

2025	Applied Generative AI Specialization <i>Michigan Engineering</i>
2020 – 2022	Fundamentals of Data Science <i>Institute for Computational & Mathematical Engineering, Stanford University</i>
2020	Data Science and Big Data Analytics: Making Data-Driven Decisions <i>MIT xPRO</i>
2017	Deep Learning Nanodegree Foundation <i>Udacity Nanodegree Program</i>
2017	Neural Network and Deep Learning <i>deeplearning.ai by Andrew Ng</i>
2016	Python Data Structure <i>University of Michigan</i>
2015	Pattern Discovery in Data Mining <i>University of Illinois at Urbana-Champaign</i>

AWARDS & FELLOWSHIPS

2025	Ruth Allen Endowment Award, Michigan State University, East Lansing, MI
2015	Graduate Office Fellowship, Michigan State University, East Lansing, MI
2015	Research Enhancement Award, Michigan State University, East Lansing, MI
2012 – 2015	Research Practicum/Development Fellowship, Michigan State University, East Lansing, MI

2011 – 2012	Summer Research Renewable Fellowship, Michigan State University, East Lansing, MI
2009	Government Scholarship for Studying Abroad, Ministry of Education, Taiwan
2001 – 2004	Academic Excellence Awards, National Kaohsiung Normal University, Taiwan
2003	Ching-Tien Educational Research Scholarship, NCKU, Taiwan
2002	His-Chen Hsu Scholarship, National Kaohsiung Normal University, Taiwan
2001	Shih-Chin Tsen Memorial Scholarship, National Kaohsiung Normal University, Taiwan

PUBLICATIONS

† Medical students who have worked directly with Dr. Chang.

Ph.D. students who have worked directly with Dr. Chang.

* Dr. Chang is the senior author.

Books

- Glick, D., Bergin, J., & **Chang, C.** (Eds.). (2023). *Supporting Self-regulated Learning and Student Success in Online Courses*. IGI Global, Hershey, Pennsylvania. <https://doi.org/10.4018/978-1-6684-6500-4>
- Glick, D., Cohen, A., & **Chang, C.** (Eds.). (2020) *Early Warning Systems and Targeted Interventions for Student Success in Online Courses*. IGI Global, Hershey, Pennsylvania.

Journal Articles

- ³³ Lee, E., **Chang, C.**, Wu, Y.-H., Hu, E. A., Myers, A., Delzell, E., Kushida, C., Lu, Y., Shi Y., Huang, S., Wang, P., Zhu, S., Hsing, A. W. (Accepted). Sleep Quality and Psychological Resilience: A Community-Based Study. Submitted to Sleep Health. *BMJ Open*.
- ³² Sawatsky, A. & **Chang, C.** (2026). Assess ‘til They Don’t Aggress: Impacting LGBTQ+ Healthcare Microaggressions with Inclusive-Skills Assessment. *Academic Medicine*.
- ³¹ Shi, Y., Li, X., Hao, Y., Wu, Q., Yu, Y., Li, S., Xu, N., Wu, Y.-H., Lee, E., **Chang, C.**, Hu, E., Lu, Y., Delzell, E., Hsing, A. W., Hu, S. (2026). Frequent sweetened beverage consumption is associated with accelerated biological aging: Evidence from a population-based study and gut microbiota analysis. *Advanced Science*, e76460. <https://doi.org/10.1002/advs.76460> [5-year impact factor: 14.8]
- ³⁰ Oostema, J., **Chang, C.**, & Reeves, M. (2026). Abstract A134: Leveraging emergency medical services narrative text to understand contributors to missed stroke recognition in the field. *Stroke*, 57(Suppl_1). https://doi.org/10.1161/str.57.suppl_1.A134
- ²⁹ Zhao, B., Li, Y., Li, S., Xu, N., Shi, Y., Yu, Y., Zheng, W., Wu, YH., Hu, E., Huang, PK., Lee, E., **Chang, C.**, Lu, Y., John, EM., Hsing, AW., He, W., Zhu, S. (2026). Associations of dietary diversity with Type 2 Diabetes in two Chinese cohorts: A multi-omics study. *Food & Function*. May 21. doi: 10.1039/d6fo01294a. Epub ahead of print. PMID: 42171301. [Impact Factor 2025: 6.3]
- ²⁸ **Chang, C.**, Laird-Fick, H., Mitchell, J., Parker, C., Solomon, D. (2025). Assessing the impact of clerkships on the growth of clinical knowledge. *Annals of Medicine*. 57(1), 2443812. <https://doi.org/10.1080/07853890.2024.2443812> [Impact Factor 2023: 4.9]
- ²⁷ Umucu, E., Lee, B., **Chang, C.**, Tsai, J. (2025). Housing status, mental health, and help-seeking among American Indian and Alaska Native Veterans in the Midwest. *Frontiers in Public Health*, 13, 1613224. [Impact Factor 2023: 3.4]

- ²⁶ Jaki, T., **Chang, C.**, Kuhlemeier, A., & Van Horn, L. (2025). Predicting individual treatment effects: Challenges and opportunities for machine learning and artificial intelligence. *KI - Künstliche Intelligenz. Special Issue: Artificial Intelligence in Medicine*. [German Journal of Artificial Intelligence - Organ des Fachbereichs "Künstliche Intelligenz" der Gesellschaft für Informatik e.V.] [Impact factor 2022: 2.9]
- ²⁵ Lee, B., **Chang, C.**, Chun, J., Lee, G., Niemiec, R., & Umucu, E. (2024). Character strengths and employment in persons with disabilities: A National Survey Study. *The Journal of Positive Psychology*. [5-year Impact Factor 2023: 4.6]
- ²⁴ Umucu, E., Lee, B., **Chang, C.**, Szymkowiak, D., & Tsai, J. (2024). Geographic and psychosocial correlates of homelessness or unstable housing among U.S. Veterans in the Midwest. *American Journal of Community Psychology*. [Impact factor 2020: 3.55]
- ²³ Ghiasi, M. †#, **Chang, C.**, Shafrir, A. L., Vitonis, A. F., Sasamoto, N., Vazquez, A. I., DiVasta, A. D., Upson, K., Sieberg, C. B., Terry, K. L., Holzman, C. B., Missmer, S. A. (2024). Subgroups of pelvic pain are differentially associated with endometriosis and inflammatory comorbidities: a latent class analysis. *Pain*, 165(9), 2119–2129. <https://doi.org/10.1097/j.pain.0000000000003218>. [Impact factor 2022: 7.4]
- ²² Zaluzec, E. K., Kenyon, E., Volk, M., Hayat, H., Powell, K., Loomis, A., Chakravarty, S., Hix, J. ML., Schipper, J., **Chang, C.**, Kiupel, M., Wang, P., Shapiro, E., Sempere, L. F. (2024). Tantalum oxide nanoparticles as versatile and high-resolution X-ray contrast agent for intraductal image-guided ablative procedure in rodent models of breast cancer. *npj Imaging*, 2(3). [Impact factor 2022: 15.2] <https://doi.org/10.1038/s44303-024-00007-5>
- ²¹ Keyes, C. W.†, van Holsbeck, B., Kerver, J. M., & **Chang, C.*** (2023). An Assessment of Knee Flexion in Lateral Knee X-rays. *Medical Student Research Journal*, <https://doi.org/10.15404/msrj/02.2023.239>
- ²⁰ **Chang, C.** (2023). How can educators prepare for the age of infodemiology? *Journal of Education Research*, 346, 72-85. <https://doi.org/10.53106/168063602023020346005>. [In Chinese]
- ¹⁹ Zhou, G., Fichorova, R. N., Holzman, C., Chen, B., **Chang, C.**, Kasten, E., & Hoffmann, H. M. (2023). Placental circadian lincRNAs and spontaneous preterm birth. *Frontiers in Genetics*. [Impact factor 2022: 4.274]
- ¹⁸ **Chang, C.**, Parker, C., Malinowski, R., Laird-Fick, H., & Gold, J. (2022). Evaluation of the peer comments for small group discussion in curriculum with virtual and in-person settings using latent space social network models. *Academic Medicine*, 97(11s):p S159. <https://doi.org/10.1097/ACM.0000000000004822> [Impact factor 2021: 6.893]
- ¹⁷ **Chang, C.**, Colón-Berlinger, M., & Mavis, B. (2022). In Reply to Yates. *Academic Medicine*, 97(10), 1430–1431. <https://doi.org/10.1097/ACM.0000000000004880> [Impact factor 2021: 6.893]
- ¹⁶ Kuhlemeier, A., Jaki, T., Jimenez, E. Y., Kong, A. S., Gill, H., **Chang, C.**, Resnicow, K., Wilson, D. K., & Van Horn, M. L. (2022). Individual differences in the effects of the ACTION-PAC intervention: an application of personalized medicine in the prevention and treatment of obesity. *Journal of Behavioral Medicine*. Advance online publication. <https://doi.org/10.1007/s10865-021-00274-2>. [Impact factor 2016: 2.5]
- ¹⁵ Kim, Y., Jeon, S., **Chang, C.**, & Chung, W. (2022) *glca*: An R Package for Multiple-Group Latent Class Analysis. *Applied Psychological Measurement*. Advance online publication.

<https://doi.org/10.1177/01466216221084197> [Impact factor 2017: 0.923]

- ¹⁴ **Chang, C.**, Jaki, T., Sadiq, M. S., Kuhlemeier, A. A., Feaster, D., Cole, N., Lamont, A., Oberski, D., & Van Horn, M. L. (2021). A permutation test for assessing the presence of individual differences in treatment effects. *Statistical Methods in Medical Research*, 30(11), 2369-2381. [Impact factor 2018: 2.388]
- ¹³ **Chang, C.***, Colón-Berlinger*, M., Mavis, B., Laird-Fick, H., Parker, C., & Solomon, D. (2021). Medical student progress examination performance and its relationship with metacognition, critical thinking, and self-regulated learning strategies. *Academic Medicine*, 96(2), 278-284. doi: 10.1097/ACM.00000000000003766. *Co-first authors, alphabetically ordered. [Impact factor 2020: 6.893]
- ¹² Kuhlemeier, A., Desai, Y., Tonigan, A., Witkiewitz, K., Jaki, T., Hsiao, Y.-Y., **Chang, C.**, & Van Horn, M. L. (2021). Applying Methods for Personalized Medicine to the Treatment of Alcohol Use Disorder. *Journal of Consulting and Clinical Psychology*. [Impact factor 2017: 4.536]
- ¹¹ **Chang, C.**, Gardiner, J., Houang, R., & Yu, Y.-L. (2020). Comparing multiple statistical software for multiple-indicator, multiple-cause modeling: An application of gender disparity in adult cognitive functioning using MIDUS II Dataset. *BMC Medical Research Methodology*. 20, 275. <https://doi.org/10.1186/s12874-020-01150-4>. [Impact factor 2019-2020: 3.412]
- ¹⁰ Zheng, B., **Chang, C.**, Lin, C.-H., & Zhang, Y. (2020). Self-efficacy, academic motivation, and self-regulation: How do they predict academic achievement for medical students? *Medical Science Educator*. <https://doi.org/10.1007/s40670-020-01143-4> [Impact factor 2019-2020: 0.668]
- ⁹ Laird-Fick, H., **Chang, C.**, Wang, L., Parker, C. J., Malinowski, R., Emery, M., & Solomon, D. (2020). Implementation of a progress clinical skills examination program in a shared discovery medical school curriculum. *Peer Journal*. doi: 10.7717/peerj.9091. [Impact factor 2019: 2.38]
- ⁸ Jaki, T., Kim, M., Lamont, A., George, M., **Chang, C.**, Feaster, D., & Van Horn, M. L. (2019). The effects of sample size on the estimation of regression mixture models. *Educational and Psychological Measurement*. 79(2). doi: 10.1177/0013164418791673. [Impact factor 2017: 1.663]
- ⁷ **Chang, C.** (2018). Zero-inflated data analysis in R. *Journal of Education Research*. 289, 69-81. doi: 10.3966/168063602018060290008. [In Chinese]
- ⁶ **Chang, C.** (2017). Application of nonparametric multilevel latent class analysis to Opportunity to Learn (OTL) classification of future mathematics teachers in the TEDS-M dataset. *Journal of Education Research*. 273, 98-111. doi: 10.3966/168063602017010273008. [In Chinese]
- ⁵ Zheng, B., Warschauer, M., Lin, C.-H., & **Chang, C.** (2016). Learning in one-to-one laptop environments: A meta-analysis and research synthesis. *Review of Educational Research*. doi:10.3102/0034654316628645. [Impact factor 2017: 8.241]
- ⁴ Raykov, T., Marcoulides, G. A., & **Chang, C.** (2016). Examining population heterogeneity in finite mixture settings using latent variable modeling. *Structural Equation Modeling: A Multidisciplinary Journal*. doi: 10.1080/10705511.2015.1103193 [Impact factor 2017: 3.531]
- ³ Raykov, T., Marcoulides, G. A., Lee, C.-L., & **Chang, C.** (2013). Studying differential item functioning via latent variable modeling: A note on a multiple testing procedure. *Educational and Psychological Measurement*. 73(5), 898-908. [Impact factor 2013: 1.43]
- ² Raykov, T., Marcoulides, G. A., Lee, C.-L., & **Chang, C.** (2013). A commentary on the relationship between model fit and saturated path models in structural equation modeling applications.

Educational and Psychological Measurement. 73(6), 1054-1068. [Impact factor 2013: 1.43]

¹ Zheng, G. Y., Liu, Y. W., **Chang, C.**, Jen, C. T., Yen, W. C., & Lin, T. H. (2005). Female students' sense of convenience of campus space at National Kaohsiung Normal University. *Journal of South Geographic*, 8, 105-125. [In Chinese]

Journal Articles – Conditional Accept, Under Review, Under Revision, or Under Preparation

Chang, C., Bao, W., Zhuang, H., Chen, J., Kong, Y., & Laird-Fick, H. (Submitted) *Automating Feedback in Medical Education: Video-Based Detection of Clinical Examination Behaviors in OSCEs*. JMIR Medical Education.

Santarossa, S., Redding, A., Wilson, C. P., Murphy, D., Sajdak, L., Alhawli, A., Bennett, M., Caldwell, M., Clemmons-Lloyd, K., Cummings, K., Doyle, J., Kesavan, S., Mitchell, V., Murphy, A., Riley, D., Stechison, L., Kippen, K., Hamilton, C., & **Chang, C.** (Submitted). Development and Psychometric Evaluation of the Patient Engagement In Research Scale enhanced by Inclusion, Diversity, Equity, Acceptance, and Safety (PEIRS-IDEAS). *Research Involvement and Engagement*

Brown, C. D., **Chang, C.**, Gardiner, J., Horton, T. H., Pfeiffer, K. A., Pearson, A. L. (Submitted). Perceived forms of gentrification are longitudinally associated with higher stress in the city of Detroit, MI.

Chang, C., Lin, J., Wu, E.-S., Wu, Y.-H., Lee, E., Min, Y., Zhao, X., Lu, Y., Lounsbury, D. W., Heaney, C. A. (Under Review). A multidimensional measure of well-being in China: Cross-sectional psychometric validation and measurement invariance of the well-being scale in WELL China. Submitted to *BMC Public Health*.

Lautman, Z., **Chang, C.**, Rangan, E., Pan, T., Hittle, M., Uwakwe, C., Kumar, S., Rader, H., & Snyder, M. P. (Minor Revision). *An Interpretable framework for cohort-level phenotyping of wearable data*. Submitted to *IEEE Journal of Biomedical and Health Informatics*.

Chang, C., Thomas J., Van Horn, L. *PITE: An R Package for Predicted Individual Treatment Effect*. Submitted to Journal of the American Medical Informatics Association.

Wu., J., **Chang, C.**, Akaeze, H., Novelli, C. *Impact of Afterschool Programs on High School Graduation and Dropout Rates: A Multiyear Michigan 21st CCLC Study*. Educational Evaluation and Policy Analysis.

Chang, C., Keyes, C.†, Laird-Fick, H., Parker, C., Wexler, B.†, & Solomon, D. *Exploring medical students' learning trajectories on subject performance and their relationships with the USMLE Step1 Score: A medical school with chief complaint-based integrated curriculum*.

Chang, C., Kelly, K., Van Horn, M. L., Houang, R. T., Gardiner, J., Van Egeren, L., & Wu, H. C. *Classification accuracy and parameter estimation in multilevel contexts: A study of conditional nonparametric multilevel latent class analysis*. Preprint: <https://arxiv.org/abs/2010.15368v2>

Chang, C., Laird-Fick, H., & Parker, C. *Assessing medical students' longitudinal history-gathering skills in Objective Structured Clinical Examination and its reliability: A testlet response model approach*.

Chang, C., Akaeze, H.#, & Van Horn, M. L. *The cluster effect on robustness of the parameter estimation in multilevel regression mixture models: A Monte Carlo Study*.

Technical Reports

Chang, C. (2019). *Conducting a simulation study using R and Mplus: An example of latent class analysis*

under 24 conditions. Technical Report: CC190417. Office of Medical Education Research and Development, College of Human Medicine, Michigan State University. Retrieved from https://github.com/quantlover/Conducting-a-Simulation-Study-Using-R-and-Mplus/blob/master/LCALabelSwitching_CC190417.pdf

Chang, C. (2015). *Project Writing Wave III Statistical Report*. East Lansing, MI: The Center for Statistical Training and Consulting.

Chang, C. (2014). *Project Writing Wave II Statistical Report*. East Lansing, MI: The Center for Statistical Training and Consulting.

Chang, C. (2014). *Project Writing Wave I Statistical Report*. East Lansing, MI: The Center for Statistical Training and Consulting.

Newsletter

Chang, C. (2008). *Prepare Teaching Chinese as a Second Language (TCSL) Teachers to Work Abroad*. No. 293. Retrieved from http://epaper.edu.tw/old/epaper.aspx?period_num=293

Press Releases

Chang, C. (2008). Certified Teaching Chinese as a Second Language (TCSL) Teachers Recruitment for Michigan and Ohio [Press release]. Retrieved from http://depart.moe.edu.tw/ED2500/News_Content.aspx?n=7EE69A276CA727DC&sms=3146F7AB2B7E0B2F&s=11EBD5DC691B53CF [in Chinese]

Chang, C. (2008). Certified Teaching Chinese as a Second Language (TCSL) Teachers Recruitment for AP Chinese course in Midwest USA [Press release]. Retrieved from http://www.edu.tw/bicer/news.aspx?news_sn=1861 [in Chinese]

Chang, C. (2007) Certified Mathematics, Science, and TCSL Teachers Recruitment for Clark District in Nevada [Press release]. Retrieved from <http://ws.moe.edu.tw/001/Upload/1/relfile/6643/10367/3b01cb19-b863-414b-8763-0aa4ea793c4b.pdf> [in Chinese]

GRANTS

Funded

¹⁹ Psychometric Consultant (PI: S. Santarossa). *Adapting and Testing a Novel Method of Engagement: Virtual Body Mapping*. Patient-Centered Outcomes Research Institute. 8/1/2026 – 9/30/2027. (Total: \$664,948, Direct: \$500,000, Indirect: \$164,948). \$2,500, 10 hours over one year.

¹⁸ Principal Investigator. *Examining Fairness in AI-Based Clinical Assessments*. President Grant. Josiah Macy Jr. Foundation. 3/1/2026 – 2/28/2027. (Total: 25,000). 10% summer, 0.30 summer months.

¹⁷ Principal Investigator. *Fair and Equitable AI Assessment: Operationalizing Measurement Invariance in OSCE Video Analytics*. Emerging Innovators Grant, National Board of Medical Examiners. 11/1/2025 – 10/31/2026. (Total: \$25,000) 3% effort. Declined to allow continued review of another NBME proposal (Stemmler Grant).

¹⁶ Co-Investigator (PI: J. Oostema) *Understanding EMS decision-making and its contribution to missed and delayed diagnosis of acute stroke in the pre-hospital setting*. AcademyHealth: Research on Pre-Hospital Diagnostic Delay. 10/15/2023 – 10/14/2024. (Total: \$150,000) 26% effort (3.1 calendar

months).

- ¹⁵ Co-Principal Investigator (PI: E. Umuru). *Homeless Veteran Evaluation*. Veteran Affairs Battle Creek. 11/1/2023 – 9/30/2024. (Total: \$200,000, Direct: \$49,880.03, Unrecovered Indirect: \$12,883.42) 35% effort AY, 33.33% Summer (3.5 person-months AY, 1.05 person-months Summer).
- ¹⁴ Co-Investigator (PI: H.-J. Wu) *Evaluation of Michigan 21st Century Community Learning Center*. Michigan Department of Education. 07/01/2020 – 06/30/2025. (Total: \$2,430,000). 30% effort (2.25 person-months AY, 0.3 person-months summer) in Year 3.
- ¹³ Co-Investigator (PI: O. Alshaarawy) *Cannabis use and cardiovascular health outcomes among patients with high blood pressure*. National Institutes of Health. 07/01/2023 – 06/30/2028. 0.18 AY and 0.06 summer months in Year 1 (2% effort, 0.24 person-months) and 0.9 AY and 0.3 summer months (10% effort, 1.2 person-months) in Year 2 – Year 5.
- ¹² Psychometric Consultant (PI: S. Santarossa). *Patient Engaged Validation of the PEIR Survey Enhanced by Inclusion, Diversity, Equity, Acceptance, and Safety (PIER - IDEAS)*. Patient-Centered Outcomes Research Institute. 8/1/2025 – 9/30/2026. (Total: \$664,948, Direct: \$500,000, Indirect: \$164,948). \$2,500, 10 hours through one year.
- ¹¹ Site PI (PI: S. Santarossa). *Patient Engaged Validation of the PEIR Survey Enhanced by Inclusion, Diversity, Equity, Acceptance, and Safety (PIER - IDEAS)*. Patient-Centered Outcomes Research Institute. 9/1/2024 – 10/31/2025. (Total: \$664,948, Direct: \$500,000, Indirect: \$164,948). 13% effort (1.43 academic months, 0.39 summer months) over 14 months.
- ¹⁰ Psychometric Consultant (PI: S. Santarossa). *Patient Engaged Validation of the PEIR Survey Enhanced by Inclusion, Diversity, Equity, Acceptance, and Safety (PIER - IDEAS)*. Patient-Centered Outcomes Research Institute. 8/1/2023 – 7/31/2025. (Total: \$664,948, Direct: \$500,000, Indirect: \$164,948). \$6,500, 25 hours through two years.
- ⁹ Principal Investigator. *Developing Speech-to-Text Models for Medical Undergraduates' Clinical Competency in Objective Structured Clinical Examinations Using Natural Language Processing*. National Board of Medical Examiners Innovation Grant. 1/1/2023 – 12/31/2023. (Total: \$25,000, Direct: \$22,727, Indirect: \$2,273) 4.2% effort (0.38 person-months AY, 0.13 person-months summer). No cost extension 1/1/2024 – 12/31/2024.
- ⁸ Principal Investigator (PI: C. Chang, Co-PI: H. Laird-Fick) *Using Computer Vision Techniques to Identify Medical Students' Safety Behaviors in Objective Structured Clinical Examinations*. American Medical Association Accelerating Change in Medical Education Innovation Grant. 9/1/2022 – 8/31/2023. (Total: \$30,000, Direct: \$30,000). 10% effort (0.9 person-months AY, 0.3 person-months summer). No cost extension 9/1/2023 – 8/31/2024.
- ⁷ Co-Principal Investigator (PI: E. Umuru). *VINC13 Homeless Veteran Evaluation*. Veteran Affairs Battle Creek. 10/1/2022 – 5/15/2023. (Total: \$49,880.03, Direct: \$49,880.03, Unrecovered Indirect: \$12,883.42) 65% effort (7.48 person-months AY).
- ⁶ Statistical Consultant (PI: W. Ma, Co-PI: G. Xu) *General Cognitive Diagnosis Model: Development, Estimation, and Applications*. National Science Foundation. 6/1/2022 – 5/31/2025. (Total: \$360,000. \$5,000 in Year 3).
- ⁵ Co-Investigator (PI: L. Sempere) *Image-Guided Intraductal Ablative Procedure for Primary Prevention of Breast Cancer*. National Institutes of Health. 12/14/2021 – 11/30/2026. (Total: \$2,837,534, Direct: \$1,829,398, Indirect: \$1,008,136). 0.45 person-months AY, 0.15 person-months summer for five

years.

- ⁴ Research Mentor (PI: B. Wexler) *The Psychological & Social Impact of SARS-CoV-2 on Michigan's Rural Communities*. Health Resources & Services Administration - Rural PREP Microresearch. 06/26/2020 – 6/25/2021. (Total: \$4,000).
- ³ Statistical Consultant (PI: J. Clarke) *BEACON's Cultural Climate Audit: Creating a Toolkit for Establishing Diversity as a Core Value Across Multi-Institutional Partnerships*. Internal Grant at Center for the Study of Evolution in Action: An NSF Science and Technology Center. 03/23/2018 – 10/1/2019. (Total: \$38,375).
- ² Co-Investigator (PI: A. Pelosi, and B. Biancucci) *Development and Testing of a Novel Canine Annuloplasty Ring for Mitral Valve Repair*. Cavalier King Charles Spaniels Charitable Trust (ACKCS), Inc. Feb. 2014 – Feb. 2015. (Direct: \$12,851).
- ¹ Co-Investigator (PI: R. Black, K. Chou, M. Millerick-May, and W. Powers) *Estimation of Health Impacts of Livestock Production*. Animal Agriculture Initiative. Jul. 2010 – Dec. 2011. (Direct: \$46,606).

Under Review

External Program Evaluator. (PI: Qiban Yan). *CyberAI: MSU Cyber security training and placement program*. National Science Foundation. The full proposal was submitted on April 3rd, 2026. Site visit is scheduled for Sep. 3rd, 2026.

PI. *AI-Assisted POCUS Assessment and Coaching: A U.S.–Taiwan Proof-of-Concept Study in Competency-Based Medical Education*. Ministry of Education, Taiwan. The full proposal was submitted on July 31st, 2026.

Consultant. (PI: Ann Hsing). HRV-derived sleep and stress in relation to cardiometabolic health: Digital phenotyping for personalized health in AsA-NHPIs. Full proposal submitted on June, 5th, 2026.

Site PI (PI: J. Mitchell). *Comprehensive Point-Of-Care Ultrasound (POCUS) Training with Motion Augmented eXperience (MAX) Feedback for Special Operations Forces (SOF) Medic*. US Department of Defense. 07/01/2026 – 06/30/2027. Submitted on Nov., 14, 2025. (Total: \$55,496.00, Direct: \$35,348.00, Indirect: \$20,148.00) 5% effort (0.6 calendar months; 0.45AY and 0.15 summer months).

PEER-REVIEWED CONFERENCE PAPERS

† Undergraduate students work with Dr. Chang

* Master students work with Dr. Chang

% Medical students work with Dr. Chang

PhD students work with Dr. Chang

Accepted or Presented

⁶⁸ Liu, S., Chiba, M., Hsing, A. W., Rich, T., **Chang, C.**, Wu, Y.-H., Tamakoshi, A., Saito, E., Tachimori, H., Ota, E. (2026). *Japanese version of the Stanford WELL for Life Scale: Translation and cultural adaptation*. Poster was accepted to present at the 2026 Yonsei International Nursing Conference, Japan.

- ⁶⁷ Lee, E., **Chang, C.**, Wu, Y.-H., Hu, E. A., Myers, A., Delzell, E., Kushida, C., Heaney, C., Huang, S., Wang, P., Lu, Y., Zhu, S., Hsing, A. W. (2026). *Sleep quality and psychological resilience across borders: Evidence from the WELL US and China Cohorts*. Poster was accepted to present at the 2026 Society for Epidemiologic Research Conference, June 23 – 26, 2026, in Phoenix, Arizona.
- ⁶⁶ Wu, Y.-H., Huang, H.-P., Hsu, C. F., Ahn, A. C., Huang, P.-K., **Chang, C.**, Lee, E., Lautman, Z., Kushida, C. A., Hsing, A. W. (2026). *Associations between heart rate variability from wearable devices and cardiovascular risk factors*. Poster was accepted to present at the 2026 Society for Epidemiologic Research Conference, June 23 – 26, 2026, in Phoenix, Arizona.
- ⁶⁵ Liu, S., Myers, A., Chiba, M., **Chang, C.**, Lee, E., Rich, T., Tamakoshi, A., Saito, E., Tachimori, H., Hsing, A., Ota, E. (2026). *Developing a psychometrically validated well-being survey: WELL Japan*. Stanford Global and Planetary Health Research Convening.
- ⁶⁴ **Chang, C.**, Vipul, J.*, Laird-Fick, H. (2025). *Natural Language Processing for Corpus and Keyword Analysis in Medical Education: Insights from Annotated OSCE Transcripts*. Paper was accepted to present at the Association of Medical Education in Europe Annual Meeting, Barcelona, Spain (Aug 2025). Withdrawn due to funding cut.
- ⁶³ **Chang, C.**, Vipul, J.*, & Laird-Fick, H. (2025). *Evaluating Annotation Performance of GPT-4o Mini, RoBERTa, and Human Annotators*. Paper was presented at the National Council of Measurement in Education, Denver, CO (Apr. 2025).
- ⁶² **Chang, C.**, Bao, W.[#], Kong Y., & Laird-Fick, H. (2025). *Ensuring Measurement Invariance and Fairness in OSCEs Using AI-Based Action Detection Models for Hand-Washing and Glove-Wearing: A Fairness Approach for Small Sample Sizes*. Paper was presented at the American Educational Research Association, Denver, CO (Apr. 2025).
- ⁶¹ **Chang, C.**, Bao, W.[#], Kong Y., & Laird-Fick, H. (2024). *Assessing Medical Students' Hand-Washing and Glove-Wearing Actions in an Objective Structured Clinical Examination Using Computer Vision*. Paper was presented at the American Association of Medical Colleges, Atlanta, GA (Nov., 2024).
- ⁶⁰ **Chang, C.** (2024). *A Systematic Review of Standard Setting Approaches in Clinical Skills Assessment for Medical Education: Opportunities and Challenges*. Paper was presented at the International Association for Health Professions Education, Basel, Switzerland (Aug, 2024).
- ⁵⁹ **Chang, C.** (2024). *Artificial Intelligence Applications in Objective Structured Clinical Examinations*. Paper was presented at the American Educational Research Association, Philadelphia, PA (Apr. 2024).
- ⁵⁸ **Chang, C.** & Sun, T. (2024). *Psychometric Considerations for Mitigating Assessment Bias*. Paper was presented at the American Educational Research Association, Philadelphia, PA (Apr. 2024).
- ⁵⁷ **Chang, C.**, Zhuang, H. *, Kong, Y., & Laird-Fick, H. (2023). *Using Computer Vision to Assess Students' Safety Behaviors in an Objective Structured Clinical Examination*. Poster presented at the annual ChangeMedEd conference, Chicago (Sep. 2023).
- ⁵⁶ **Chang, C.** & Akaeze, H.[#] (2023). *Predicting medical students' specialty choices using geographical variables: Findings from data spanning 30 years*. Paper presented at the annual meeting of the American Educational Research Association, Chicago, IL (Apr. 2023).
- ⁵⁵ **Chang, C.**, Parker, C., Malinowski, R., & Gold, J. (2022). *Evaluation of the peer comments for small group discussion in curriculum with virtual and in-person settings*. The paper was presented at Learn Serve Lead 2022: The AAMC Annual Meeting, Nashville, TN (Nov. 2022).

- ⁵⁴ **Chang, C.,** Laird-Fick, H., & Parker, C. (2022). *Assessing medical students' longitudinal history-gathering skills in Objective Structured Clinical Examination and its reliability: A testlet response model approach*. Paper presented at the annual meeting of the American Educational Research Association, San Diego, CA (Apr. 2022).
- ⁵³ **Chang, C.,** Laird-Fick, H., & Parker, C. (2021). *Application and challenges of applying artificial intelligence in assessing undergraduate medical students*. Poster presented at the annual ChangeMedEd conference, virtual (Sep. 30, 2021).
- ⁵² McCaffery, H. & **Chang, C.** (2021). *Simulated stromatolites: Are Martian fossils likely to be found?* Paper presented at the Joint Statistical Meeting, virtual (Aug. 11, 2021).
- ⁵¹ **Chang, C.,** Debnath, I.[†], & McCaffery, H. (2021). *Evaluating the classification accuracy of five machine-learning classification methods in identifying rare cases in educational assessment settings*. Paper presented at the Joint Statistical Meeting, virtual (Aug. 11, 2021).
- ⁵⁰ Wexler, B.%, Haque, B., Keyes, C.%, Alavi, Z., & **Chang, C.** (2021). *The impact of SARS-CoV-2 on Mental Health in Michigan's Rural Communities*. Paper presented at the National Rural Health Conference, virtual (Apr. 2021).
- ⁴⁹ **Chang, C.,** Laird-Fick, H., Parker, C., Wexler, B. %, Keyes, C. W. %, & Solomon, D. (2021). *Assessing the impact of clerkships on the growth of clinical knowledge*. Paper presented at the 2021 the Central Group on Educational Affairs, virtual (Apr. 2021)
- ⁴⁸ **Chang, C.,** Laird-Fick, H., Parker, C., & Solomon, D. (2021). *Alternative Reliability Measures for Medical Students' Skills in Objective Structured Clinical Examinations*. Paper presented at the annual meeting of the American Educational Research Association, virtual (Apr. 2021).
- ⁴⁷ Parker, C., Laird-Fick, H., **Chang, C.,** Wang, L., Malinowski, R., Emery, M., & Solomon, D. (2020). *The generalizability of a progress clinical skills examination for assessing the growth in clinical skills*. 2020 Northeastern Group on Educational Affairs (NEGEA). (Conference canceled).
- ⁴⁶ **Chang, C.** (2020). *Using neural network for latent transition hierarchical cognitive diagnostic models*. The 2020 National Council on Measurement in Education, Minneapolis, MN (Conference canceled).
- ⁴⁵ **Chang, C.,** Laird-Fick, H., Parker, C., Wang, L., & Solomon, D. (2020). *What do entrustable professional activities (EPAs) measure? Evidence from exploratory and confirmatory factor analyses*. The American Educational Research Association, San Francisco, CA (Conference canceled).
- ⁴⁴ **Chang, C.,** Laird-Fick, H., Parker, C., Wang, L., & Solomon, D. (2020). *Assessing medical students' clinical performance progress and the relationship with USMLE Step 1 scores*. The American Educational Research Association, San Francisco, CA (Conference canceled).
- ⁴³ Zheng, B., **Chang, C.,** & Lin, C.-H. (2020). *Self-efficacy, academic motivation, and self-regulation: How do they predict academic achievement for medical students?* The American Educational Research Association, San Francisco, CA (Conference canceled).
- ⁴² Wang, L., Laird-Fick, H., **Chang, C.,** Parker, C., Malinowski, R., & Solomon, D. (2020). *Evaluating medical students' trajectory on progress tests using Markov Chain model*. The American Educational Research Association, San Francisco, CA (Conference canceled).
- ⁴¹ Laird-Fick, H., Solomon, D., Wang, L., **Chang, C.,** Parker, C., & Malinowski, R. (2019). *Implementation of a progress clinical skills examination program in a new medical school*

- curriculum*. Paper presented at The Generalists in Medical Education, Phoenix, AZ (Nov. 2019).
- ⁴⁰ Wang, L., Laird-Fick, H., Parker, C., **Chang, C.**, Malinowski, R., & Solomon, D. (2019). *Can a "wake-up call" improve medical students' performance in their first year?* Paper presented at the annual meeting of the Association of American Medical Colleges Group on Educational Affairs, Phoenix, AZ (Nov. 2019).
- ³⁹ Mavis, B., **Chang, C.**, Laird-Fick, H., Thompson-Busch, A., Thompson, M., Emery, M., & Wagner, D. (2019). *Comparing graduates' self-assessment with program directors' ratings of entrustability*. Paper presented at the annual meeting of the Association of American Medical Colleges Group on Educational Affairs, Phoenix, AZ (Nov. 2019).
- ³⁸ Wang, L., Laird-Fick, H., Parker, C., **Chang, C.**, Malinowski, R., & Solomon, D. (2019). *Performance of medical school students in small groups based on competency-based testing in a new medical school curriculum*. Paper presented at the annual meeting of the Association of American Medical Colleges Group on Educational Affairs, Phoenix, AZ (Nov. 2019).
- ³⁷ **Chang, C.** & McCaffery, H. (2019). *Classification accuracy of longitudinal diagnostic assessment models: A simulation study using a neural network approach*. Paper presented at the Joint Statistical Meeting, Denver, CO (Jul. 2019).
- ³⁶ McCaffery, H. & **Chang, C.** (2019). *Controlling the false discovery rate: A simulation study*. Paper presented at the Joint Statistical Meeting, Denver, CO (Jul. 2019).
- ³⁵ **Chang, C.** (2019). *Examining the performance of a neural network cognitive diagnostic model in hierarchy-structured attributes*. Paper presented at the National Council on Measurement in Education, Toronto, Canada (Apr. 2019).
- ³⁴ **Chang, C.**, Colon-Berlingeri, M., Mavis, B., Solomon, D., Wang, L., Parker, C., & Laird-Fick, H. (2019). *Examining the relationship of medical students' metacognition, critical thinking, learning strategies, and academic performance*. Paper presented at the Annual Conference of the American Educational Research Association, Toronto, Canada (Apr. 2019).
- ³³ **Chang, C.** & Mavis, B. (2019). *Examining the relationships between student background, Entrustable Professional Activities, and program directors' expectations*. Paper presented at the Central Group on Educational Affairs, Grand Rapids, MI (Mar. 2019).
- ³² Wang, L., Laird-Fick, H., Parker, C., **Chang, C.**, Malinowski, R., & Solomon, D. (2019). *Association between USMLE Step 1 test performance with competency-based testing*. Paper presented at the Central Group on Educational Affairs, Grand Rapids, MI (Mar. 2019).
- ³¹ Wang, L., Laird-Fick, H., Parker, C., **Chang, C.**, Malinowski, R., & Solomon, D. (2019). *Performance of medical school students in small groups based on competency-based testing in a new medical school curriculum*. Paper presented at the Central Group on Educational Affairs, Grand Rapids, MI (Mar. 2019).
- ³⁰ **Chang, C.**, Solomon, D., Parker, C., Malinowski, R., Wang, L., & Laird-Fick, H. (2018). *Identifying subtypes of medical students' performance assessment results using a latent class quadratic growth analysis (LCQGA)*. Paper presented at The Generalists in Medical Education, Austin, TX (Nov. 2018).
- ²⁹ **Chang, C.**, Solomon, D., Parker, C., Wang, L., Malinowski, R., & Laird-Fick, H. (2018). *Examining medical students' domain performance using time-varying longitudinal data*. Paper presented at The Generalists in Medical Education, Austin, TX (Nov. 2018).

- ²⁸ Wang, L., Solomon, D., Parker, C., Laird-Fick, H., **Chang, C.**, & Malinowski, R. *Improvement in Essential Basic Science Knowledge over the First Two Years of Medical School in an Innovative Competency-Based Curriculum*. Paper presented at The Generalists in Medical Education, Austin, TX (Nov. 2018).
- ²⁷ **Chang, C.** (2018). *Classification Accuracy of Unsupervised Learning Methods with Discrete and Mixture Distributed Indicators: A Monte Carlo Simulation Study*. Paper presented at the Joint Statistical Meetings, Vancouver, Canada (Jul. 2018).
- ²⁶ **Chang, C.** & Van Horn, M. L. (2018). *Cluster effects on parameter estimation in multilevel regression mixture models*. Paper presented at the Annual Conference of the Modern Modeling Methods Conference, Storrs, CT (May 2018).
- ²⁵ **Chang, C.** & Tong, X. (2017). *Nonparametric multilevel latent class analysis: An approach to classification in multilevel contexts*. Paper presented at National Council on Measurement in Education 2014 Annual Meeting, San Antonio, TX (Apr. 2017).
- ²⁴ **Chang, C.** & Tong, X. (2017). *A simulation study of parameter estimates of nonparametric multilevel latent class analysis with covariates*. Paper presented at the Annual Conference of the American Educational Research Association, San Antonio, TX (Apr. 2017).
- ²³ Li, T., Jackson-Elmoore, C., Akaeze, H., **Chang, C.**, & Lawrence, F. (2016). *Modeling changes in legislators' information source usage in the policy process*. Paper presented at the Annual Meeting of the American Political Science Association, Philadelphia, PA (Sep. 2016).
- ²² **Chang, C.**, Jackson-Elmoore, C., Li, T., Akaeze, H., & Lawrence, F. (2016). *Information source usage by American state legislators*. Paper presented at the Annual Conference of the European Political Science Association, Brussels, Belgium (Jun. 2016).
- ²¹ **Chang, C.** (2016). *Nonparametric multilevel latent class analysis: A comparative study of teachers' beliefs about teaching and learning in 24 countries*. Paper presented at the Annual Conference of the Modern Modeling Methods Conference, Storrs, CT (May 2016).
- ²⁰ **Chang, C.**, Maier, K. S., Houang, R. T., Van Egeren, L., & Wu, H-J. (2016). *Multilevel latent class analysis (MLCA) as an alternative method for program evaluation: An application using data from the Michigan 21st Century Community Learning Centers Grant Program*. Paper presented at the Annual Conference of the American Educational Research Association, Washington, DC (Apr. 2016).
- ¹⁹ Zheng, B., **Chang, C.**, & Chen, C.-H. (2016). *A meta-analysis of one-to-one computing on academic achievements*. Paper presented at the Annual Conference of American Educational Research Association, Washington, DC (Apr. 2016).
- ¹⁸ **Chang, C.** (2015). *Sensitivity analysis for multivalued treatment effect: An example of a cross-country study of teacher participation and job satisfaction*. Paper presented at the 2015 Biannual Meeting of the Society for Research on Educational Effectiveness, Washington, DC (Mar. 2015).
- ¹⁷ **Chang, C.**, Houang, R. T., & Maier, K. S. (2015). *The power analysis for two-level hierarchical models with treatment at cluster level in non-randomized design*. Paper presented at the Annual Conference of American Educational Research Association, Chicago, IL (Apr. 2015).
- ¹⁶ **Chang, C.** & Mao, L. (2014). *The effect of termination criteria on cognitive diagnostic computerized adaptive testing*. Paper presented at National Council on Measurement in Education 2014 Annual Meeting, Philadelphia, PA (Apr. 2014).
- ¹⁵ Kao, S.-C., Zara, T., Woo, A., & **Chang, C.** (2013). *Analysis of ability changes for repeating examinees*

- using latent growth curve models. Paper presented at the 2nd Annual Conference on the Statistical Detection of Potential Test Fraud, Madison, WI (Oct. 2013).
- ¹⁴ Cheng, C. & **Chang, C.** (2013). *The relationship among gender, culture and math abilities: A perspective from mediation analysis*. Paper presented at the Annual Conference of American Educational Research Association, San Francisco, CA (Apr. 2013).
- ¹³ Tong, B. & **Chang, C.** (2013). *An investigation of factors affecting students' mathematics growth*. Paper presented to 2013 NCME Graduate Student Issues Committee (GSIC) Research Session, San Francisco, CA (Apr. 2013).
- ¹² **Chang, C.** & Tong, B. (2013). *How professional development can do better for teachers, and who needs it? The implications from hierarchical logistic regression analysis and latent class analysis*. Paper presented at the 2013 Biannual Meeting of the Society for Research on Educational Effectiveness, Washington, DC (Mar. 2013).
- ¹¹ **Chang, C.** (2012). *A longitudinal study of class size and mathematics achievement among immigrant and non-immigrant groups*. Paper presented at the 120th Annual Conference of the American Psychological Association, Orlando, FL (Aug. 2012).
- ¹⁰ Jani, V., **Chang, C.**, Hussain, S., Kassab, M., Fitzsimmons, B.-F., & Zaidat, O. (2012a). *The triglyceride: High-density lipoprotein-cholesterol ratio in prediction of development of atherosclerotic disease process*. Paper presented at the 64th Annual Meeting of the American Academy of Neurology, New Orleans, LA (Apr. 2012).
- ⁹ Jani, V., **Chang, C.**, Hussain, S., Kassab, M., Fitzsimmons, B.-F., & Zaidat, O. (2012b). *Association of inflammatory markers in vascular death: A retrospective overview*. Paper presented at the 64th Annual Meeting of the American Academy of Neurology, New Orleans, LA (Apr. 2012).
- ⁸ **Chang, C.** & Maier, K. S. (2012). *Power analysis for two-level hierarchical models with propensity score applied to different levels*. Paper presented at the 2012 Biannual Meeting of the Society of Research on Educational Effectiveness, Washington, DC (Mar. 2012).
- ⁷ **Chang, C.** (2011). *Teacher's beliefs, attitudes, and professional development: A cross-country analysis using multilevel modeling*. Paper presented at the Annual Meeting of the International Meeting of the Psychometric Society, Hong Kong (Jul. 2011).
- ⁶ **Chang, C.** (2011). *Tendency, decision-making processes, and satisfaction of international students in Taiwan*. Paper presented at the Annual Meeting of the Comparative and International Education Society, Montreal, Canada (May 2011).
- ⁵ Steward, R. J., **Chang, C.**, Chen, T., & Tong, B. (2010a). *In search of optimal levels of counseling competence: A study of counselor trainees' backgrounds on skill acquisition*. Paper presented at the Annual Meeting of the American Psychology Association, Division 45, Ann Arbor, MI (Jun. 2010).
- ⁴ Steward, R. J., **Chang, C.**, Chen, T., & Tong, B. (2010b). *Predicting exceptional status among counselor trainees with origins beyond the USA*. Paper presented at the Annual Meeting of the American Psychology Association, Division 17 Great Lake Conference, Akron, OH (Apr. 2010).
- ³ Steward, R. J., **Chang, C.**, Ali, K., Riojas, P., Chen, H., & Neal, S. (2010). *Identifying predictors of high school ranking: An exploratory study of the influence of school, student and district*. Paper presented at the Annual Meeting of the American Psychology Association, Division 17 Great Lake Conference, Akron, OH (Apr. 2010).
- ² **Chang, C.** (2007). *A comparison study of Chinese as a Second Language Policies in China and Taiwan*.

Hawaii International Conference on Education, Honolulu, HI (Jan. 2007).

¹ **Chang, C.** (2006). *A comparison study of attracting international students policies in USA, Japan, France, Germany, and Australia*. National Chi-Nan University, Nantou, Taiwan (Aug. 2006).

TEACHING AND MENTORING

Course Director

Department of Statistics and Probability, Michigan State University

2020 STT 422: Statistics II

Department of Epidemiology and Biostatistics, Michigan State University

2018 – 2021, 2026 EPI 851: SAS Programming I Essentials

2019 – 2022 EPI 852: SAS Programming II Project Management

Department of Individual, Family, and Community Education, The University of New Mexico

2017 IFCE 511: Introductory Educational Statistics

Preceptor

College of Human Medicine, Michigan State University

2019 – 2021 HM556 Intersession: Evidence-based Medicine (Epidemiology)

Rotation Study Groups (RSGs)

College of Human Medicine, Michigan State University

2027 HM564 Epidemiology

Guest Lecturer

Measurement and Quantitative Methods Program, Michigan State University

2018 CEP 923: Item Response Theory

Department of Individual, Family, and Community Education, University of New Mexico

2017 IFCE 603: Applied Statistical Design and Analysis

Invited Speech & Workshop

Taiwan Health Information Society Association in American Medical Information Association (THISA - AMIA)

2026 Precision Medical Education: A Measurement Science Perspective

Department of Nutrition & Dietetics and Human Development & Family Studies, Central Michigan University

2016 Multilevel Modeling

The Armed Forces Medical College, Egypt

2021 Hands-on Assessment and Examination Analysis

National Chengchi University, Taiwan

- 2026 Beyond Statistical Models: Causal Inference and Machine Learning in Educational Research
- 2025 Statistical Inference and Educational Decision-Making: Simulation, Bayesian Methods, and Policy Recommendation

National Kaohsiung Normal University, Taiwan

- 2025 Can AI Understand Empathy? Applications of Affective AI in Education

University of California, San Francisco

- 2025 Grand Round: AI-assisted applications in OSCE assessment: Bridging the Gap in Clinical Evaluation

Teaching Assistantship

Department of Counseling, Educational Psychology, and Special Education, Michigan State University

- 2011, 2013 CEP 920: Instrument Construction in Education
- 2015 CEP 921: Psychometric Theory I
- 2012, 2013, 2016 CEP 932: Quantitative Methods in Educational Research I
- 2012, 2013, 2015 CEP 933: Quantitative Methods in Educational Research II
- 2011, 2013 CEP 934: Multivariate Data Analysis I

Department of Education, National Chengchi University, Taiwan

- 2006 – 2007 Time Series Analysis Applied in Education
- 2005 – 2006 Behavior Modification
- 2004 – 2005 School Architecture and Equipment
- 2003 – present Private Statistical Consultant, Statistical Tutoring

Mentoring

Post-Doc, Stanford Prevention Research Center, Stanford Medicine

- 2025 – present Lee, Eunhye. *Hsing's lab*,

Doctoral Dissertation Working Group, Department of Bioengineering, Stanford University

- 2023 – present Lautman, Ziv (Expected PhD Fall 2025). *Engineering Wellness*

Doctoral Dissertation Committee, Department of Epidemiology and Biostatistics, Michigan State University

Advisor

- 2022 Summer Xia, Kun (Expected Biostatistics PhD Fall 2026).

Member

- 2022 – present Glazier, Alicynne (Expected PhD Spring 2024).

2020 – 2021 Ghiasi, Marzieh (Expected PhD Fall 2021). *Genetic heterogeneity associated with phenotypic variation in clinical presentation of endometriosis*

Doctoral Dissertation Committee, Department of Geography, Michigan State University

2024 – present Katie Brown (Expected PhD Fall 2025).

Doctoral Dissertation Committee, Measurement and Quantitative Methods Program, Michigan State University

2022 – present Li, Tatyana. *Logistic Regression under Class Imbalance*

2021 – present Sullivan, William (Expected PhD Spring 2026).

2018 – 2020 Akaeze, Hope O. *Incorporating Differential Speed in Cognitive Diagnostic Models with Polytomous Attributes*

Doctoral Practicum Paper Committee, Measurement and Quantitative Methods Program, Michigan State University

2016 Akaeze, Hope O. *Sensitivity of multicollinearity-induced bias to outcome-covariate relationship*

2016 Kim, Sewon. *A comparison of sequential and stratified two-stage sampling designs for populations with the unknown distribution of sampling units*

2015 Li, Tenglong. *Applying H-H and H-T estimators to propensity score analysis*

Master's Thesis Advisor, Department of Epidemiology and Biostatistics, Michigan State University

2021 – 2022 Zheng, Zhuochen. *Comparing multiple software for latent class analysis: Determinants for adults' executive functions.*

2019 – 2021 Debnath, Ishita. *Evaluating the prediction accuracy of multinomial logistic regression analysis with sparsity: A simulation study*

Research Mentor, MD program, College of Human Medicine, Michigan State University

2020 – 2023 Wexler, Benjamin

2020 – 2023 Keyes, Christopher

Academic Guidance Committee, Measurement and Quantitative Methods Program, Michigan State University

2024 – present Subir Hait

ADVANCED TRAINING

2020 *Translational Bioinformatics in R*, Michigan State University

2020 *Statistical Learning for Process Data (log files)*, Columbia University

2019 *Social Media Mining in R: Collecting and Analyzing Twitter Data*, Michigan State University

2019 *Natural Language Processing 1 + 2*

2016 *Programming for Everybody (Getting Started with Python)* by the University of

PROFESSIONAL SERVICE

Chief Operating Officer

2025– present Medical Social Network, *Non-profit organization*.

Guest Journal Editor

2024 Promoting Longitudinal Assessment in Medical Education. *Medical Education Online*

Assistant Journal Editor

2006 *Journal of Education Research* (vol.136.), Taiwan

Ad-hoc Grant Review Panel

2019 Edward J. Stemmler Medical Education Research Fund,
National Board of Medical Examiners

Grant Review Panel

2021 National Science Foundation
2021 American Education Research Association Division I Grant Program

Review Committee

2018 – present American Psychological Association, Dissertation Research Award

Journal Reviewer

- Perspectives on Medical Education
- Psychological Medicine
- Structural Equation Modeling: A Multidisciplinary Journal
- Educational Measurement: Issues and Practice
- Academic Medicine
- Advances in Health Sciences Education
- Journal of Teacher Education
- Thinking Skills and Creativity
- BMC Open
- Technology in Society
- Humanities and Social Sciences Communications

Ad-hoc Journal Reviewer

- Teaching and Learning in Medicine: An International Journal

Conference Reviewer

- Association of American Medical Colleges
- American Education Research Association
- National Council on Measurement in Education
- Central Group on Educational Affairs

Conference Coordinator

- 1st International Conference on Creativity and Innovation Education

National-level committee

2024 – present The Research in Medical Education (RIME) Program Planning Committee,
Association of American Medical Colleges (AAMC) Medical Education Meeting

State-level Committee

Slide Judge Committee – Representing Michigan State University College of Human Medicine

2022, 2023, 2024 Southeast Michigan Center for Medical Education (SEMCME) Research Forum
and Annual Michigan Quality Improvement Summit

College-level Committee

Center for Statistical Training and Consulting, Michigan State University

2026 Unit Review Committee for Promotion Evaluation

College of Human Medicine, Michigan State University

2018 – present Assessment Analytics Team
2021 – 2024 Student Competence Committee
2022 – 2024 Liaison Committee for Medical Education (LCME) Faculty subcommittee
2024 – 2025 Faculty Search Committee, Doug Meijer Medical Innovation Building Global
Impact Initiative

Doug Meijer Medical Innovation Building Global Impact Initiative

2024 – 2025 Faculty Search Committee, College of Human Medicine, Michigan State University

Department-level Committee

Reappointment, Promotion, and Tenure Committee

2023 - Committee member, Office of Medical Education Research and Development,
College of Human Medicine, Michigan State University

Recruitment Committee

2020 – 2021 Department of Epidemiology and Biostatistics, College of Human Medicine,
Michigan State University

Award Committee

2020 – 2022 Chair, Office of Medical Education Research and Development,

College of Human Medicine, Michigan State University

2020 – 2022 Committee member, Office of Medical Education Research and Development,
College of Human Medicine, Michigan State University

Faculty Advisory Committee

2021 – present Office of Medical Education Research and Development,
College of Human Medicine, Michigan State University

Search Committee

2026 Office of Medical Education Research and Development
Fixed-term assistant professor Search Committee Chair, College of Human
Medicine, Michigan State University

2020 Office of Medical Education Research and Development
Tenure-stream assistant professor Search Committee Member, College of Human
Medicine, Michigan State University

Other Universities Service

2025 External Reviewer for Tenure promotion, Department of Medical Education,
University of Illinois College of Medicine

2024 External Reviewer for Fixed Term promotion, Rutgers School of Health Professions

WORK EXPERIENCE

2012 – 2016 Statistical Consultant, The Center for Statistical Training and Consulting (CSTAT),
Michigan State University, East Lansing, MI
Supervisor: Dr. Steve Pierce

2011 – 2016 Teaching Assistant, Measurement and Quantitative Methods Program,
Michigan State University, East Lansing, MI
Supervisor: Dr. Kimberly Kelly and Dr. Tenko Raykov

2010 – 2011 Research Assistant, The Center for Mathematics and Science Education Research,
Michigan State University, East Lansing, MI
Supervisor: Dr. Richard Houang

2019 – 2010 Research Assistant, Department of Physical Medicine and Rehabilitation,
Michigan State University, East Lansing, MI
Supervisor: Dr. Michael Leahy and Dr. Virginia Thielsen

2007 – 2009 Staff, Bureau of International Cultural and Educational Relations,
Ministry of Education, Taipei, Taiwan
Supervisor: Dr. Chin-Jen Liu

2005 – 2007 Research Assistant and Teaching Assistant, National Chengchi University, Taipei, Taiwan
Supervisor: Dr. Hsen-Hsing Ma

2004 – 2005 Research Assistant, National Chengchi University, Taipei, Taiwan
Supervisor: Dr. Chih-Min Tang

2003 – 2004 Research Assistant, National Kaohsiung Normal University, Kaohsiung, Taiwan
Supervisor: Dr. Shu-Mei Chang

SKILLS

Measurement and Statistical Software

- R, Python, SAS, SPSS, *Mplus*, LatentGOLD, IRTPro, Multilog, ArcGIS, LISREL, HLM, Frontier, AHP, Stata

Language

- | | |
|----------------------|----------------------|
| • Taiwanese – native | • English – fluent |
| • Chinese – native | • Spanish – beginner |

REPERTOIRE of RESEARCH METHODS

- | | |
|---|--|
| • Multivariate Data Analysis | • Meta-Analysis |
| • Multilevel Models | • Bayesian data analysis |
| • Instrument Construction in Education | • Statistical Computing |
| • Psychometric Theory | • Clinical Trials |
| • Item Response Theory | • Duration and Severity Analysis |
| • Multidimensional Item Response Theory | • Biostatistics Models in Genomic Analysis |
| • Computer Adaptive Testing | • Time Series Analysis in Educational Research |
| • Differential Item Functioning | • Missing Data |
| • Scale Development | • Power Analysis & Sample Size Planning |
| • Categorical Data Analysis | • Survey Research |
| • Advanced Measurement | • Experimental Design |
| • Longitudinal Data Analysis | • Program Evaluation and Assessment |
| • Latent Variable Modeling | • Evidence-Based Medicine |
| • Causal Inference | |